

Лабораторная работа №16

Администрирование локальных сетей

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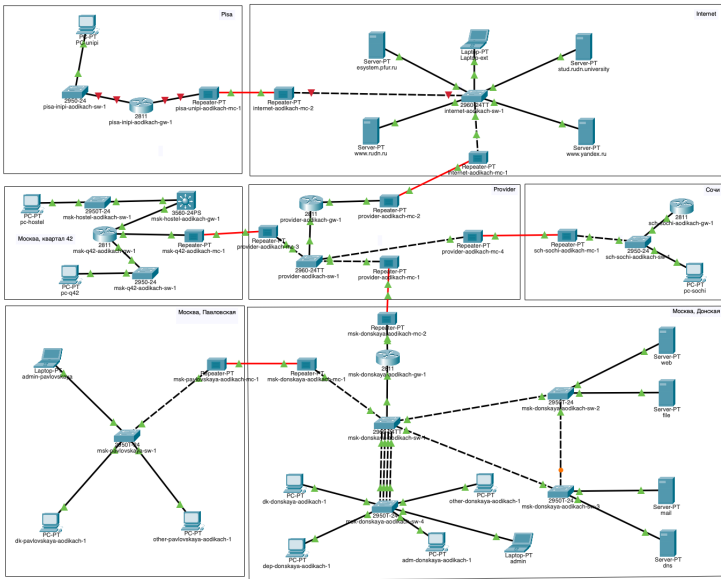
Информация

- Дикач Анна Олеговна
- НПИбд-01-22 (1132222009)
- Российский университет дружбы народов
- <https://github.com/chelibos?tab=repositories>

Получение навыков настройки VPN-туннеля через незащищённое Интернет соединение.

Выполнение лабораторной работы

Размещаю в рабочей области проекта оборудование для улицы г. Пиза



В физической чатси создаю новый город Пиза, улицу и переносу всё необходимое оборудование

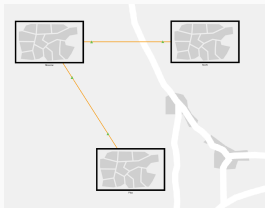
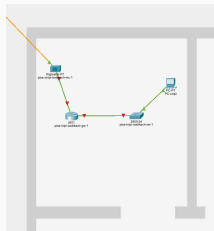


Рис. 2: Создание нового города



Провожу первоначальную настройку площадки города Пиза. Для этого настраиваю маршрутизатор и коммутатор

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname pisa-inipi-aodikach-gw-1
pisa-inipi-aodikach-gw-1(config)#
pisa-inipi-aodikach-gw-1(config)#line vty 0 4
pisa-inipi-aodikach-gw-1(config-line)#password cisco
pisa-inipi-aodikach-gw-1(config-line)#login
pisa-inipi-aodikach-gw-1(config-line)#exit
pisa-inipi-aodikach-gw-1(config)#line console 0
pisa-inipi-aodikach-gw-1(config)#line console 0
^
% Invalid input detected at '^' marker.

pisa-inipi-aodikach-gw-1(config)#line console 0
pisa-inipi-aodikach-gw-1(config-line)#password cisco
pisa-inipi-aodikach-gw-1(config-line)#login
pisa-inipi-aodikach-gw-1(config-line)#exit
pisa-inipi-aodikach-gw-1(config)#enable secret cisco
pisa-inipi-aodikach-gw-1(config)#service password-encryption
pisa-inipi-aodikach-gw-1(config)#username admin privilege 1 secret cisco
pisa-inipi-aodikach-gw-1(config)#ip domain-name unipi.edu
pisa-inipi-aodikach-gw-1(config)#crypto key generate rsa
pisa-inipi-aodikach-gw-1(config)#crypto key generate rsa
^
% Invalid input detected at '^' marker.

pisa-inipi-aodikach-gw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-inipi-aodikach-gw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-inipi-aodikach-gw-1(config)#line vty 0 4
*Mar 1 0:12:53.884: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:12:53.884: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-inipi-aodikach-gw-1(config-line)#transport input ssh
pisa-inipi-aodikach-gw-1(config-line)#exit
pisa-inipi-aodikach-gw-1(config)#exit
pisa-inipi-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console
```



```

pisa-inipi-aodikach-sw-1(config)#
pisa-inipi-aodikach-sw-1(config)#line vty 0 4
pisa-inipi-aodikach-sw-1(config-line)#password cisco
pisa-inipi-aodikach-sw-1(config-line)#login
pisa-inipi-aodikach-sw-1(config-line)#exit
pisa-inipi-aodikach-sw-1(config)#line console 0
pisa-inipi-aodikach-sw-1(config-line)#password cisco
^
% Invalid input detected at '^' marker.

pisa-inipi-aodikach-sw-1(config-line)#password cisco
pisa-inipi-aodikach-sw-1(config-line)#login
pisa-inipi-aodikach-sw-1(config-line)#exit
pisa-inipi-aodikach-sw-1(config)#enable secret cisco
pisa-inipi-aodikach-sw-1(config)#service password-encryption
pisa-inipi-aodikach-sw-1(config)#username admin privilege 1 secret cisco
pisa-inipi-aodikach-sw-1(config)#ip domain-name unipi.edu
pisa-inipi-aodikach-sw-1(config)#crypto key generate rsa
The name for the keys will be: pisa-inipi-aodikach-sw-1.unipi.edu
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
    a few minutes.

How many bits in the modulus [512]:
% Generating 512 bit RSA keys, keys will be non-exportable...[OK]

pisa-inipi-aodikach-sw-1(config)#line vty 0 4
*Mar 1 0:15:52.248: RSA key size needs to be at least 768 bits for ssh version 2
*Mar 1 0:15:52.248: %SSH-5-ENABLED: SSH 1.5 has been enabled
pisa-inipi-aodikach-sw-1(config-line)#transport input ssh
pisa-inipi-aodikach-sw-1(config-line)#exit
pisa-inipi-aodikach-sw-1(config)#exit
pisa-inipi-aodikach-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-inipi-aodikach-sw-1#wr m
Building configuration...
[OK]

```

Рис. 5: Настройка коммутатора

Провожу настройку интерфейсов на маршрутизаторе и коммутаторе

```
-----
pisa-inipi-aodikach-gw-1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
pisa-inipi-aodikach-gw-1(config)#interface f0/0
pisa-inipi-aodikach-gw-1(config-if)#no shutdown

pisa-inipi-aodikach-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

pisa-inipi-aodikach-gw-1(config-if)#exit
pisa-inipi-aodikach-gw-1(config)#interface f0/0.401
pisa-inipi-aodikach-gw-1(config-subif)#
%LINK-5-CHANGED: Interface FastEthernet0/0.401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.401, changed state to up

pisa-inipi-aodikach-gw-1(config-subif)#encapsulation dot1Q 401
pisa-inipi-aodikach-gw-1(config-subif)#ip address 10.131.0.1 255.255.255.0
pisa-inipi-aodikach-gw-1(config-subif)#description unipi-main
pisa-inipi-aodikach-gw-1(config-subif)#exit
pisa-inipi-aodikach-gw-1(config)#interface f0/1
^
% Invalid input detected at '^' marker.

pisa-inipi-aodikach-gw-1(config)#interface f0/1
pisa-inipi-aodikach-gw-1(config-if)#no shutdown

pisa-inipi-aodikach-gw-1(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

pisa-inipi-aodikach-gw-1(config-if)#ip address 192.0.2.20 255.255.255.0
pisa-inipi-aodikach-gw-1(config-if)#description internet
pisa-inipi-aodikach-gw-1(config-if)#exit
pisa-inipi-aodikach-gw-1(config)#ip route 0.0.0.0 0.0.0.0 192.0.2.1
pisa-inipi-aodikach-gw-1(config)#exit
pisa-inipi-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-inipi-aodikach-gw-1#wr m
Building configuration...
[OK]
pisa-inipi-aodikach-gw-1#
```

```
pisa-inipi-aodikach-sw-1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
pisa-inipi-aodikach-sw-1(config)#interface f0/24
pisa-inipi-aodikach-sw-1(config-if)#switchport mode trunk
pisa-inipi-aodikach-sw-1(config-if)#exit
pisa-inipi-aodikach-sw-1(config)#interface f0/1
pisa-inipi-aodikach-sw-1(config-if)#switchport mode access
pisa-inipi-aodikach-sw-1(config-if)#switchport access vlan 401
% Access VLAN does not exist. Creating vlan 401
pisa-inipi-aodikach-sw-1(config-if)#exit
pisa-inipi-aodikach-sw-1(config)#vlan 401
pisa-inipi-aodikach-sw-1(config-vlan)#name unipi-main
pisa-inipi-aodikach-sw-1(config-vlan)#exit
pisa-inipi-aodikach-sw-1(config)#interface vlan401
pisa-inipi-aodikach-sw-1(config-if)#
%LINK-5-CHANGED: Interface Vlan401, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan401, changed state to up

pisa-inipi-aodikach-sw-1(config-if)#no shutdown
pisa-inipi-aodikach-sw-1(config-if)#exit
pisa-inipi-aodikach-sw-1(config)#
pisa-inipi-aodikach-sw-1(config)#exit
pisa-inipi-aodikach-sw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-inipi-aodikach-sw-1#wr m
Building configuration...
[OK]
pisa-inipi-aodikach-sw-1#
```

Рис. 7: Настройка интерфейсов коммутатора

Далее перехожу к настройке VPN на основе GRE на маршрутизаторе на улице Донская и маршрутизатора в городе Пиза

```
msk-donskaya-aodikach-gw-01#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-aodikach-gw-01(config)#interface Tunnel0

msk-donskaya-aodikach-gw-01(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

msk-donskaya-aodikach-gw-01(config-if)#ip address 10.128.255.253 255.255.255.252
msk-donskaya-aodikach-gw-01(config-if)#tunnel source f0/1.4
msk-donskaya-aodikach-gw-01(config-if)#tunnel desription 192.0.2.20
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-aodikach-gw-01(config-if)#tunnel description 192.0.2.20
                                     ^
% Invalid input detected at '^' marker.

msk-donskaya-aodikach-gw-01(config-if)#tunnel destination 192.0.2.20
msk-donskaya-aodikach-gw-01(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

msk-donskaya-aodikach-gw-01(config-if)#exit
msk-donskaya-aodikach-gw-01(config)#interface loopback0

msk-donskaya-aodikach-gw-01(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

msk-donskaya-aodikach-gw-01(config-if)#ip address 10.128.254.1 255.255.255.255
msk-donskaya-aodikach-gw-01(config-if)#exit
msk-donskaya-aodikach-gw-01(config)#ip routr 10.128.254.5 255.255.255.255 10.128.255.254
                                     ^
% Invalid input detected at '^' marker.
```

```

pisa-inipi-aodikach-gw-1#conf
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line. End with CNTL/Z.
pisa-inipi-aodikach-gw-1(config)#interface Tunnel0

pisa-inipi-aodikach-gw-1(config-if)#
%LINK-5-CHANGED: Interface Tunnel0, changed state to up

pisa-inipi-aodikach-gw-1(config-if)#ip address 10.128.255.254 255.255.255.252
pisa-inipi-aodikach-gw-1(config-if)#tunnel source f0/1
pisa-inipi-aodikach-gw-1(config-if)#tunnel destination 198.51.100.2
pisa-inipi-aodikach-gw-1(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up

pisa-inipi-aodikach-gw-1(config-if)#exit
pisa-inipi-aodikach-gw-1(config)#interface loopback0

pisa-inipi-aodikach-gw-1(config-if)#
%LINK-5-CHANGED: Interface Loopback0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback0, changed state to up

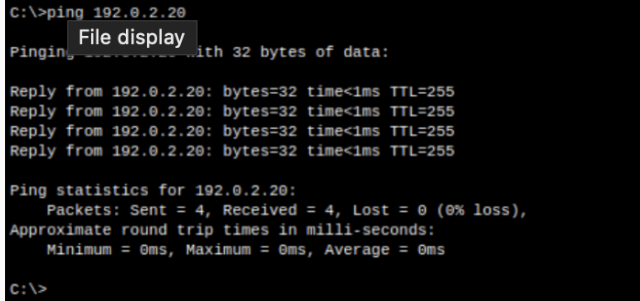
pisa-inipi-aodikach-gw-1(config-if)#ip address 10.128.254.5 255.255.255.255
pisa-inipi-aodikach-gw-1(config-if)#exit
pisa-inipi-aodikach-gw-1(config)#ip route 10.128.254.1 255.255.255.255 10.128.255.253
pisa-inipi-aodikach-gw-1(config)#router ospf 1
pisa-inipi-aodikach-gw-1(config-router)#router-id 10.128.254.5
pisa-inipi-aodikach-gw-1(config-router)#network 10.0.0.0 0.255.255.255 area 0
pisa-inipi-aodikach-gw-1(config-router)#exit
pisa-inipi-aodikach-gw-1(config)#
pisa-inipi-aodikach-gw-1(config)#exit
pisa-inipi-aodikach-gw-1#
%SYS-5-CONFIG_I: Configured from console by console

pisa-inipi-aodikach-gw-1#wr m
Building configuration...
[OK]
pisa-inipi-aodikach-gw-1#

```

Рис. 9: Настройка pisa-unipi-aodikach-gw-1

Проверяю корректность выполнения с помощью пингования администратором сети Донская узлов университета в городе Пиза



```
C:\>ping 192.0.2.20
Pinging 192.0.2.20 with 32 bytes of data:

Reply from 192.0.2.20: bytes=32 time<1ms TTL=255
Reply from 192.0.2.20: bytes=32 time<1ms TTL=255
Reply from 192.0.2.20: bytes=32 time<1ms TTL=255
Reply from 192.0.2.20: bytes=32 time<1ms TTL=255

Ping statistics for 192.0.2.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Рис. 10: Проверка корректности

Получила навыки настройки VPN-туннеля через незащищённое Интернет соединение.